

Assignment 1: Process Investigation

```
vboxuser@ubuntuabse:~$ sleep 100 &
[1] 3487
vboxuser@ubuntuabse:~$ sleep 200 &
[2] 3488
vboxuser@ubuntuabse:~$ sleep 300 &
[3] 3489
vboxuser@ubuntuabse:~$ jobs
[1]  Running                sleep 100 &
[2]-  Running                sleep 200 &
[3]+  Running                sleep 300 &
vboxuser@ubuntuabse:~$ yes > /dev/null &
[4] 3490
vboxuser@ubuntuabse:~$ yes > /dev/null &
[5] 3491
vboxuser@ubuntuabse:~$ jobs
[1]  Running                sleep 100 &
[2]  Running                sleep 200 &
[3]  Running                sleep 300 &
[4]-  Running                yes > /dev/null &
[5]+  Running                yes > /dev/null &
vboxuser@ubuntuabse:~$ ps aux | egrep "(sleep)"
vboxuser    3488  0.0  0.0   8360  2228 pts/0    S   13:03   0:00  sleep 200
vboxuser    3489  0.0  0.0   8360  2228 pts/0    S   13:03   0:00  sleep 300
vboxuser    3517  0.0  0.0   9364  2436 pts/0    S+  13:06   0:00  grep -E --color=auto (sleep)
[1] Done                  sleep 100
vboxuser@ubuntuabse:~$ top
```

```
top - 13:07:05 up 6 min,  2 users,  load average: 1.87, 1.12, 0.53
Tasks: 235 total,  3 running, 232 sleeping,  0 stopped,  0 zombie
%Cpu(s): 13.3 us, 27.3 sy,  0.0 ni, 59.3 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem : 5418.3 total, 3326.2 free, 1223.5 used, 1114.7 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 4194.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3490	vboxuser	20	0	8360	2156	2040	R	100.0	0.0	2:59.47	yes
3491	vboxuser	20	0	8360	2156	2040	R	99.7	0.0	2:43.11	yes
2360	vboxuser	20	0	292948	101376	66080	S	3.0	1.8	0:05.93	Xorg
2651	vboxuser	20	0	4790468	417452	154404	S	2.0	7.5	0:11.67	gnome-shell
3372	vboxuser	20	0	848820	55988	44880	S	1.0	1.0	0:01.42	gnome-terminal-
321	systemd+	20	0	18140	7808	6872	S	0.3	0.1	0:00.17	systemd-oomd
1	root	20	0	25296	15980	10916	S	0.0	0.3	0:01.44	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.01	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool workqueue release

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31 root      -51  0      0      0      0 S  0.0  0.0  0:00.00 idle_inject/2
32 root      rt   0      0      0      0 S  0.0  0.0  0:00.55 migration/2
[2] Done                                sleep 200
vboxuser@ubuntuabse:~$ pstree -p $$
bash(3383)─pstree(3536)
           └─sleep(3489)
              └─yes(3490)
                 └─yes(3491)
vboxuser@ubuntuabse:~$ lsof | grep yes
yes          3490          vboxuser  cwd      DIR           8,2      4096 2093058 /home/vboxuser
yes          3490          vboxuser  rtd      DIR           8,2      4096      2 /
yes          3490          vboxuser  txt      REG           8,2    35288 2617744 /usr/bin/yes
yes          3490          vboxuser  mem      REG           8,2   5726160 2617815 /usr/lib/locale/l
ocale-archive
yes          3490          vboxuser  mem      REG           8,2   2178688 2635301 /usr/lib/x86_64-l
inux-gnu/libc.so.6
yes          3490          vboxuser  mem      REG           8,2        613 264485 /usr/share/locale
-langpack/en/LC_MESSAGES/coreutils.mo
yes          3490          vboxuser  mem      REG           8,2   251544 2635298 /usr/lib/x86_64-l
inux-gnu/ld-linux-x86-64.so.2
yes          3490          vboxuser  0u       CHR        136,0      0t0      3 /dev/pts/0
yes          3490          vboxuser  1w       CHR         1,3      0t0      5 /dev/null
yes          3490          vboxuser  2u       CHR        136,0      0t0      3 /dev/pts/0
yes          3491          vboxuser  cwd      DIR           8,2      4096 2093058 /home/vboxuser
yes          3491          vboxuser  rtd      DIR           8,2      4096      2 /
yes          3491          vboxuser  txt      REG           8,2    35288 2617744 /usr/bin/yes
yes          3491          vboxuser  mem      REG           8,2   5726160 2617815 /usr/lib/locale/l
ocale-archive
yes          3491          vboxuser  mem      REG           8,2   2178688 2635301 /usr/lib/x86_64-l
inux-gnu/libc.so.6
yes          3491          vboxuser  mem      REG           8,2        613 264485 /usr/share/locale
-langpack/en/LC_MESSAGES/coreutils.mo
yes          3491          vboxuser  mem      REG           8,2   251544 2635298 /usr/lib/x86_64-l
inux-gnu/ld-linux-x86-64.so.2
yes          3491          vboxuser  0u       CHR        136,0      0t0      3 /dev/pts/0
yes          3491          vboxuser  1w       CHR         1,3      0t0      5 /dev/null
yes          3491          vboxuser  2u       CHR        136,0      0t0      3 /dev/pts/0
[3] Done                                sleep 300

```

Assignment 2: Signal Lab

```
vboxuser@ubuntuabse:~$ sleep 1000
^Z
[1]+  Stopped                  sleep 1000
vboxuser@ubuntuabse:~$ jobs
[1]+  Stopped                  sleep 1000
vboxuser@ubuntuabse:~$ ps
  PID TTY          TIME CMD
 3388 pts/0        00:00:00 bash
 3680 pts/0        00:00:00 sleep
 3681 pts/0        00:00:00 ps
vboxuser@ubuntuabse:~$ kill -CONT 3680
vboxuser@ubuntuabse:~$ jobs
[1]+  Running                  sleep 1000 &
vboxuser@ubuntuabse:~$ fg
sleep 1000
^Z
[1]+  Stopped                  sleep 1000
vboxuser@ubuntuabse:~$ kill -9 3680
vboxuser@ubuntuabse:~$ jobs
[1]+  Killed                   sleep 1000
vboxuser@ubuntuabse:~$
```

Assignment 3: PATH Hijack Experiment

```
vboxuser@ubuntuabse:~$ mkdir ~/fakebi
vboxuser@ubuntuabse:~$ echo '#!/bin/bash' > ~/fakebi/ls
vboxuser@ubuntuabse:~$ echo 'echo "Hijacked"' >> ~/fakebi/ls
vboxuser@ubuntuabse:~$ chmod +x ~/fakebi/ls
vboxuser@ubuntuabse:~$ export PATH=~/fakebi:$PATH
vboxuser@ubuntuabse:~$ ls
Hijacked
vboxuser@ubuntuabse:~$
```

```
vboxuser@ubuntuabse:~$ echo $PATH
/home/vboxuser/fakebi:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
vboxuser@ubuntuabse:~$ export PATH=$(echo $PATH | sed 's|/home/vboxuser/fakebi:||')
vboxuser@ubuntuabse:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
vboxuser@ubuntuabse:~$ ls file2.txt > stdout.txt
```